

STAT C1000 - Introduction to Statistics

GENERAL DESCRIPTION

Approval Date

October 2024

Effective Semester

Fall 2025

Department

Mathematics

Course Number

STAT C1000

Course Title

Introduction to Statistics

Course Outline Originator

Sonny Mohammadzadeh

Co-Contributor(s)

Lily Lum

Ekaterina Fuchs

Department Chairperson

Ekaterina Fuchs

Dean

David Yee

COURSE SPECIFICS

Hours

Lecture: 87.5

Homework: 175

Total Hours: 262.5

Units

5

Requisite(s)

Prerequisite: Placement as determined by the college's multiple measures assessment process or completion of a course taught at or above the level of intermediate algebra

Recommended Prep: Readiness for college-level English or ESL 188

Course Justification

This course, formerly MATH 80, meets the four-year college graduation requirement for students needing introductory college-level Statistics. It has been modified to meet the requirements of CCN. CCSF GE Area 2, Cal-GETC Area 2.

Field Trips

No

Description of Field Trips

N/A

Method of Grading

Only Letter

Repeatability

0 - Course is not repeatable

Discipline(s)

Mathematics

CATALOG DESCRIPTION

This course is an introduction to statistical thinking and processes, including methods and concepts for discovery and decision-making using data. Topics include descriptive statistics; probability and sampling distributions; statistical inference; correlation and linear regression; analysis of variance, chi-squared, and t-tests; and application of technology for statistical analysis, including the interpretation of the relevance of the statistical findings. Students apply methods and processes to applications using data from a broad range of disciplines.

STUDENT LEARNING OUTCOMES

Upon completion of this course, a student will be able to:

- A. Assess how data were collected and recognize how data collection affects what conclusions can be drawn from the data.
- B. Identify appropriate graphs and summary statistics for variables and relationships between them and correctly interpret information from graphs and summary statistics.
- C. Describe and apply probability concepts and distributions.
- D. Demonstrate an understanding of, and ability to use, basic ideas of statistical processes, including hypothesis tests and confidence interval estimation.
- E. Identify appropriate statistical techniques and use technology-based statistical analysis to describe, interpret, and communicate results.
- F. Evaluate ethical issues in statistical practice.

CONTENTS

- A. Part 1: Required topics
 1. Introduction to statistical thinking and processes
 2. Technology-based statistical analysis
 3. Applications using data from four or more of the following disciplines: administration of justice, business, economics, education, health science, information technology, life science, physical science, political science, psychology, and social science
 4. Units (subjects/cases) and variables in a data set, including multi-variable data sets
 5. Categorical and quantitative variables
 6. Sampling methods, concerns, and limitations, including bias and random variability
 7. Observational studies and experiments
 8. Data summaries, visualizations, and descriptive statistics
 9. Probability concepts
 10. Probability distributions (e.g., binomial, normal)
 11. Sampling distributions and the Central Limit Theorem

12. Estimation and confidence intervals
13. Hypothesis testing, including t-tests for one and two populations, Chi-squared test(s), and ANOVA; and interpretations of results
14. Regression, including correlation and linear regression equations

B. Part 2: Expanded topics

1. Descriptive statistics

1. Tabular representations (frequency tables and contingency tables)
2. Pictorial representations (histograms, boxplots, barcharts, and scatterplots)
3. Measures of central tendency (mean and median)
4. Measures of dispersion (standard deviation and interquartile range)
5. Other summary statistics (5-number summary, range, and outliers)
6. Use of technology to determine numerical summaries of categorical and quantitative data

2. Gathering and using data

1. Sample surveys
2. Observational studies
3. Experimental studies
4. Bias in data
5. Applications using data from other disciplines
 1. Business
 2. Social science
 3. Health science
 4. Political science
 5. Education
 6. Psychology

3. Probability

1. Sample spaces and assignment of probabilities
2. Events, including mutually exclusive events and independent events, and applications
3. Probabilities of events using techniques among the following
 - a. Basic Probability Formula for equally likely outcomes
 - b. Conditional probability
 - c. Bayes' Rule
 - d. Independence

4. Random variables

1. Probability distribution of a random variable
2. Expected value of a random variable and its meaning
3. Variance and standard deviation of a random variable and their meanings
4. Probability distributions (binomial, normal, student's t, and Chi-square)

5. Estimation

1. Basic concepts of sampling including
 - a. Random samples
 - b. Sampling with replacement
 - c. Sampling without replacement
2. Central Limit Theorem

3. Random samples and estimation of population parameters
4. The concept of interval estimation with emphasis on the following
 - a. Confidence level
 - b. Margin of error
 - c. Methods of determining sample size
5. Applications using data from other disciplines
 1. Economics
 2. Biological science
 3. Political science
 4. Education
6. Test of hypotheses
 1. Random samples to test statistical hypotheses
 - a. The mean of a single population
 - b. The proportion of a single population
 - c. The difference of means from two populations
 - d. The difference of proportions from two populations
 2. Small sample methods or large sample methods as appropriate
 3. Level of significance, Type I error, and Type II error
 4. p-value for a test
 5. Chi-square test and use technology to compute the chi-square statistic
7. Regression
 1. Random samples to estimate correlation coefficients and to compute equations of least squares regression lines
 2. Linear regression analysis and analysis of variance (ANOVA)
 3. Use of technology to determine linear correlation coefficients, least squares regression lines, and ANOVA
 4. Applications using data from other disciplines
 1. Education
 2. Health science
 3. Information technology
 4. Physical science

INSTRUCTIONAL METHODOLOGY

Assignments

1. **In-class activity:** Instructor-led discussion on topics such as descriptive statistics and gathering and using data.
2. **In-class activity:** Individualized or small group work on topics such as analyzing data and graphs; computing central tendency and dispersion, probability, and estimation for a population parameter; and performing hypothesis testing.
3. **In-class activity:** Periodic individualized or small group work using technology such as Excel, minitab, R, StatCrunch, or graphing calculator for a large dataset
4. **Out-of-class assignment:** Regular reading assignments on material covered in class such as regression.

5. **Out-of-class assignment:** Regular homework problems, for example, summarizing data using descriptive statistics (finding the mean and median, percentiles, and standard deviation); understanding the basic concepts of probability; finding expected value of a random variable, Z-scores, and confidence interval for a population parameter; performing hypothesis test; and computing correlation coefficient and linear regression lines
6. **Out-of-class assignment:** Periodic assignments using technology such as Excel, minitab, R, StatCrunch, or graphing calculator for a large dataset

Evaluation

1. **Other:** Part 1: Examples of potential methods of evaluation used to observe or measure students' achievement of course outcomes and objectives could include but are not limited to quizzes, exams, laboratory work, field journals, projects, research demonstrations, etc. Methods of evaluation are at the discretion of local faculty.
2. **Other:** Part 2: Additional methods of evaluation
3. **Participation:** Active participation in class discussions demonstrating an understanding of how data collection methods affect what conclusions can be drawn from the data as well as ethical issues that impact statistical practice.
4. **Written work:** Successful completion of written assignments in which students critique studies, analyze raw data, evaluate and interpret data distributions, apply statistical models, and demonstrate other skills related to the course content.
5. **Exams/Quizzes/Tests:** Approximately three to four tests designed to assess knowledge of sample surveys, descriptive statistics, basic principles of probability theory, random variables, application of normal distribution and Central Limit Theorem, and linear regression
6. **Final Assessment:** A comprehensive final exam on key probability and statistics areas of study, including normal distribution, estimation of parameters, tests for hypotheses, application of the Central Limit Theorem, theoretical probability, and linear regression

Representative Textbooks and Other Instructional Materials

1. Creative Commons Initiative. 2024. *Probability and Statistics*. Online Learning Initiative
2. Çetinkaya-Runde, M., Hardin, J.. 2024. *Introduction to Modern Statistics 2e*. OpenIntro
3. (Required for Common Course Numbering) Part 1: • Introduction to Modern Statistics 2e, Çetinkaya-Runde, M., Hardin, J., OpenIntro, 2024 (\$0- 25): <https://www.openintro.org/book/ims/> • Statistics: Learning From Data 3e, Peck, R., Case, C., Cengage, 2024 (\$57-250): <https://www.cengage.com/c/new-edition/9780357758298/> • Introductory Statistics: Exploring the World Through Data 4e, Gould, R., Wong, R., Ryan, C., Pearson, 2025 (\$65-80): <https://www.pearson.com/en-us/subject-catalog/p/introductory-statistics/P200000011641/9780138242145> • Introductory Statistics 2e, Illowsky, B., Dean, S., OpenStax, 2023 (\$0): <https://openstax.org/details/books/introductory-statistics-2e> • Introductory Statistics: Analyzing Data with Purpose, The Dana Center Mathematics Pathways, Charles A. Dana Center, University of Texas at Austin, 2021 (\$0): <https://www.utdanacenter.org/products/introductory-statistics>
4. A scientific or graphing calculator
5. Computer with Excel, minitab, R, StatCrunch
6. Instructor developed materials, for example, supplementary exercises or notes
7. Website: [Open Learning Initiative at Carnegie Mellon University, Probability and Statistics](https://openlearninginitiative.cmu.edu/probability-and-statistics/)

TITLE 5 CLASSIFICATION

CREDIT/DEGREE APPLICABLE (meets all standards of Title 5. Section 55002(a))